ENBC332: Statistics, Data Analysis, and Data Visualization
Fall 2025

Credits: 3
Prerequisites: Minimum grade of C- in BIOE241 or approved prior study in MATLAB

Instructor
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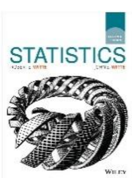
Meeting times and location
Lecture: BSE IV Room 4321, Mon/Wed 12:00-1:15 PM
Office hours: BSE IV Room 4115, Mon/Wed 1:30-2:30 PM (or email to schedule an appointment)

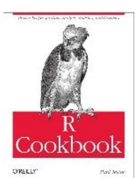
Course Description
This course will instruct you in the fundamentals of probability and statistics through examples from biological phenomena and clinical data analysis. Data visualization strategies will also be covered. Topics include **descriptive statistics**, **types of data**, **quantitative data**, **qualitative data**, **data summary**, **describing relationships**, **covariance**, **correlation coefficient**, **distributions and probability theory**, **inferential statistics**, **hypothesis testing**, **significance level**, **p-value**, **dependent or independent**, **hypothesis errors**, **simple linear regression**, **R-Square**, **ordinary least squares**, **T-test**, **F-test**, **multi-linear regression**, **adjusted R-squared**, **Analysis of Variance (ANOVA)**. These topics are explored using the statistical package R, with a focus on understanding how to use and interpret output from this programming language as well as how to visualize results.

Textbooks

Required:

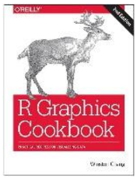


Statistics 11th Edition | eTextbook
Robert S. Witte, John S. Witte
Wiley, 2017

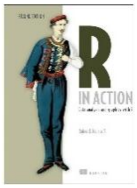


R Cookbook: Proven Recipes for Data Analysis, Statistics, and Graphics
Paul Teeor
O'Reilly Media, 2011

Recommended:



R Graphics Cookbook: Practical Recipes for Visualizing Data
Wilson Chang
O'Reilly Media, 2018



R in Action: Data Analysis and Graphics with R
Rob Kabacoff
Manning, 2015

Course Learning Outcomes

In this course, you will

- Appreciate the science of statistics and the scope of its potential applications
- Be able to summarize and present data in meaningful ways
- Be able to select the appropriate statistical analysis depending on the question at hand
- Be able to conduct, present, and interpret common statistical analyses using R
- Be able to effectively and clearly describe results from analyses performed to others

Class Policies

Lectures

- Lecture notes will be posted on **ELMS-Canvas** after each class. These notes will serve as the basis for exam material and homework (HW). You are highly encouraged to visit the Canvas site regularly to stay up-to-date with the posted materials.

Communication: Canvas or office hours ONLY

- I encourage students to speak with me during office hours or via ELMS email. If a student is not able to attend office hours, meetings by appointment can be scheduled.
- I will answer ELMS communication as soon as possible, definitely within 24 hours on a weekday or within 48 hours on the weekend. ELMS is an essential part of the course, and you should master the use of this resource.
- All questions related to the course whether content, homework, or etc. (except personal) must be posted on ELMS, so all students can read the question and answers. I will answer the questions as fast as possible. Students are encouraged to post answers to these questions, and I will comment afterwards.
- Please send me an email in case you have administrative questions or something urgent/important related to the course.

Attendance

- Attendance is highly recommended as material covered in class will be the backbone of homework and exams.
- It is highly recommended that students review the assigned textbook material before class.

Usage of Generative AI

- The use of generative AI tools (e.g., ChatGPT, Gemini, or DeepSeek) is permitted only for **learning** and **idea development purposes**, such as brainstorming or clarifying concepts.

- If generative AI tools are used for idea development, you must clearly disclose how the tools were used (e.g., "Used ChatGPT to brainstorm analysis approaches" in a comment or footnote). Failure to disclose AI assistance may be considered academic dishonesty.
- Any direct copying, paraphrasing, or closely replicating AI-generated content in assignments is strictly prohibited and will result in a grade of **ZERO**. Repeated violations may lead to formal disciplinary action.
- The use of generative AI is strictly prohibited in the exams as well. Any evidence of AI use will result in a grade of **ZERO** and be reported to the Office of Student Conduct.
- AI tools are known to generate inaccurate or fabricated information. Students are fully responsible for verifying all submitted content.
- When in doubt about acceptable use, consult the instructor in advance.

Assignments

- Homework assignments (30%) will be posted on ELMS and will be due on the date and time listed on each assignment.
- Late homework can be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Quizzes

- Quizzes (10%) will emphasize the topic given in the corresponding week.
- Expect the quizzes every **Wednesday**.
- There will be no make-up for missed quizzes. You will be given one of the quizzes as a "freebie" in case you are sick, etc.

Course evaluation

- Student feedback on lectures every **Wednesday after the quiz**.

Exams

- The midterm exam will cover all content taught before the midterm.
- The project will be assigned before the midterm.
- Project presentations will take place during the last class session.
- The final exam will be cumulative, covering material from the entire course.

Grading

30% - Assignments – [5% class discussion]
10% - Quizzes
20% - Midterm
20% - Group Project
20% - Final Exam

Grading Scale

The instructor will use a "+/–" grading scheme for final grades.

Grade	**% Attained**
A−, A, A+	90 – 100
B−, B, B+	80 – 89
C−, C, C+	70 – 79
D−, D, D+	60 – 69
F	< 60

Important Note: This syllabus, along with course assignments and due dates, are subject to change. It is the student's responsibility to check ELMS for corrections or updates to the syllabus.

Tips on How to Do Well

- **Come to lecture, pay attention/take notes, and do the assigned readings** *A lot of content will be covered in this class and it will be very hard to cram the material the night before an exam. The easiest way to make sure you are staying up-to-date with the material is to simply come to class and do the reading during the week of the lecture.*
- **Do the assignments**. *The assignments will give you an idea of what the exam will look like. Do them on time.*
- **Form a study group.** *Many new concepts will be covered before each exam. Form a study group to discuss the material and test one another. Work together to make sure you have a comprehensive understanding of the material.*

Accommodations for Students with Disabilities: The University is required to provide appropriate accommodations for students with disabilities. Students with disabilities should inform instructors of their needs at the beginning of the semester so that we can implement appropriate accommodations. The Accessibility and Disability Service (ADS: a division of the Counseling Center) stands ready to assist students and faculty in determining and implementing appropriate academic recommendations. ADS will work closely with both faculty and students. You may contact the office at 301-314-7682. For more information, please visit http://www.counseling.umd.edu/ads/.

Honor Code: The University of Maryland has a nationally recognized Code of Academic Integrity, administered by the Student Honor Council. This Code sets standards for academic integrity in Maryland for all undergraduate and graduate students. As a student, you are responsible for upholding these standards for this course. It is very important for you to be aware of the consequences of cheating, fabrication, facilitation, and plagiarism. For more information on the Code of Academic Integrity or the Student Honor Council, please visit http://www.studenthonorcouncil.umd.edu/whatis.html. In particular, the Code of Academic Integrity prohibits students from cheating on exams, plagiarizing papers, submitting the same paper for credit in two courses without authorization, buying papers, submitting fraudulent documents, and forging signatures. The University also has adopted an Honor Pledge that should be written and signed on all submitted papers. The Pledge reads: I pledge on my honor that I have not given or received any unauthorized assistance on this assignment/examination.

Policy on Religious Holidays: The University System of Maryland policy on religious observances provides that "students should not be penalized because of observances of their religious beliefs; students shall be given an opportunity, whenever feasible, to make within a reasonable time any academic assignment that is missed due to individual participation in religious observances." However, "it is the student's responsibility to inform the instructor of any intended absences for religious observances in advance. Notice should be provided as soon as possible but no later than the end of the schedule adjustment period."

UMD Course Related Policies: For further information about the University of Maryland policies for undergraduate students, please visit http://www.ugst.umd.edu/courserelatedpolicies.html

Schedule (subject to change)

Week	Date	Topic
1	Labor Day September 3	**Course Overview** **Introduction:** Descriptive Statistics, Inferential Statistics, R Programming
2	September 8 September 10	**Types of Data, Survey vs. Experiment, Population vs. Sample** **Descriptive Statistics: Data Summary:** Measures of Location and Variability
3	September 15 September 17	**Probability, Distributions and Probability Theory:** Normal Distribution, Standard Normal Distribution Probability
4	September 22 September 24	Probability Density Function, Cumulative Distribution Function, Cumulative Distribution Function, Binomial Distribution, Central Limit Theorem
5	September 29 October 1	**Describing Relationships:** Scatterplot, Covariance, Correlation Coefficient, Correlation vs. Causation
6	October 6 October 8	**Research in Biocomputational Engineering:** Instructor or Guest lecturer **Review for Midterm Exam**
7	October 13 October 15	**Fall Break** **MIDTERM Exam – Everything covered so far**
8	October 20 October 22	**Inferential Statistics:** Point Estimates, Confidence Intervals, T-distribution **Stating Hypotheses:** Hypothesis Testing, Significance level
9	October 27 October 29	p-value, Hypothesis Errors **Dependent or Independent:** Dependent Sample (Paired Samples)
10	November 3 November 5	**Dependent or Independent:** Independent Sample (Unpaired Samples) for known population variances and unknown population variances
11	November 10 November 12	**Regression Analysis:** Simple Linear Regression (SLR), R-squared, SLR tests
12	November 17 November 19	**Regression diagnostics:** Linear Regression Assumptions, Omitted Variable, Outliers, Dummy Variables
13	November 24 November 26	Multi-Linear Regression (MLR), Adjusted R-squared, F-test for MLR
14	December 1 December 3	**Analysis of Variance (ANOVA):** One-Way ANOVA **Review for Final Exam: All materials covered in this course**
15	December 8 December 10	**Review for Final Exam: All materials covered in this course** **Project presentations**
	TBD	**FINAL EXAM – All course materials**